



US 20250282024A1

(19) **United States**(12) **Patent Application Publication**  
**KUBODERA et al.**(10) **Pub. No.: US 2025/0282024 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **GRINDING TOOL DRESSING METHOD AND GRINDING DEVICE****Publication Classification**(51) **Int. Cl.****B24B 53/017** (2012.01)**B24B 53/085** (2006.01)(52) **U.S. Cl.**CPC ..... **B24B 53/017** (2013.01); **B24B 53/085** (2013.01)(71) Applicant: **HONDA MOTOR CO., LTD.,**  
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**ABSTRACT**

A method for dressing a grinding tool includes acquiring information on a deflection waveform indicating a positional deviation of a first tooth surface for each rotation phase of a first rotating body, acquiring, as contact positional data indicating a position where the first tooth surface and the second tooth surface are in contact with each other, an envelope of the deflection waveform, the envelope passing through a plurality of crests of the deflection waveform located on a side closer to a second tooth surface, and dressing a grinding tooth surface with a dresser tooth surface by changing the rotation speed of the first rotating body based on a predetermined forming quantity and the contact positional data.

(21) Appl. No.: **19/058,515**(22) Filed: **Feb. 20, 2025**(30) **Foreign Application Priority Data**

Mar. 11, 2024 (JP) ..... 2024-037060

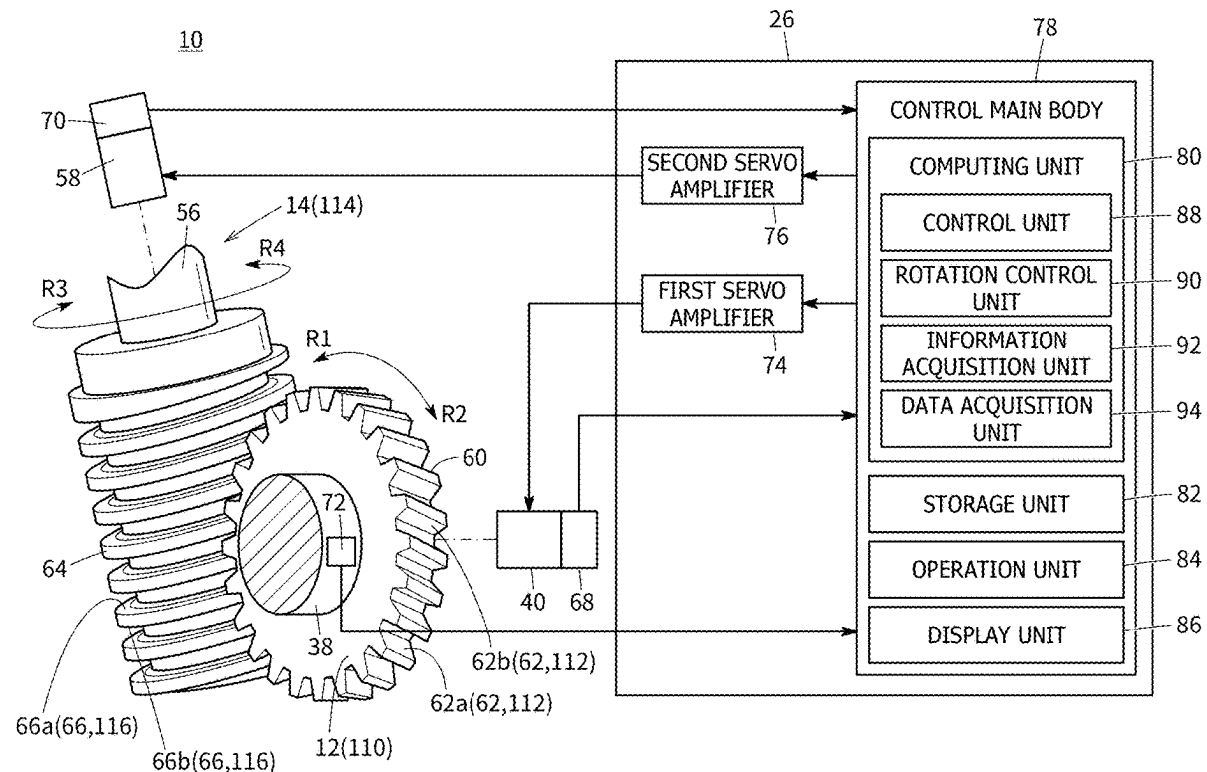
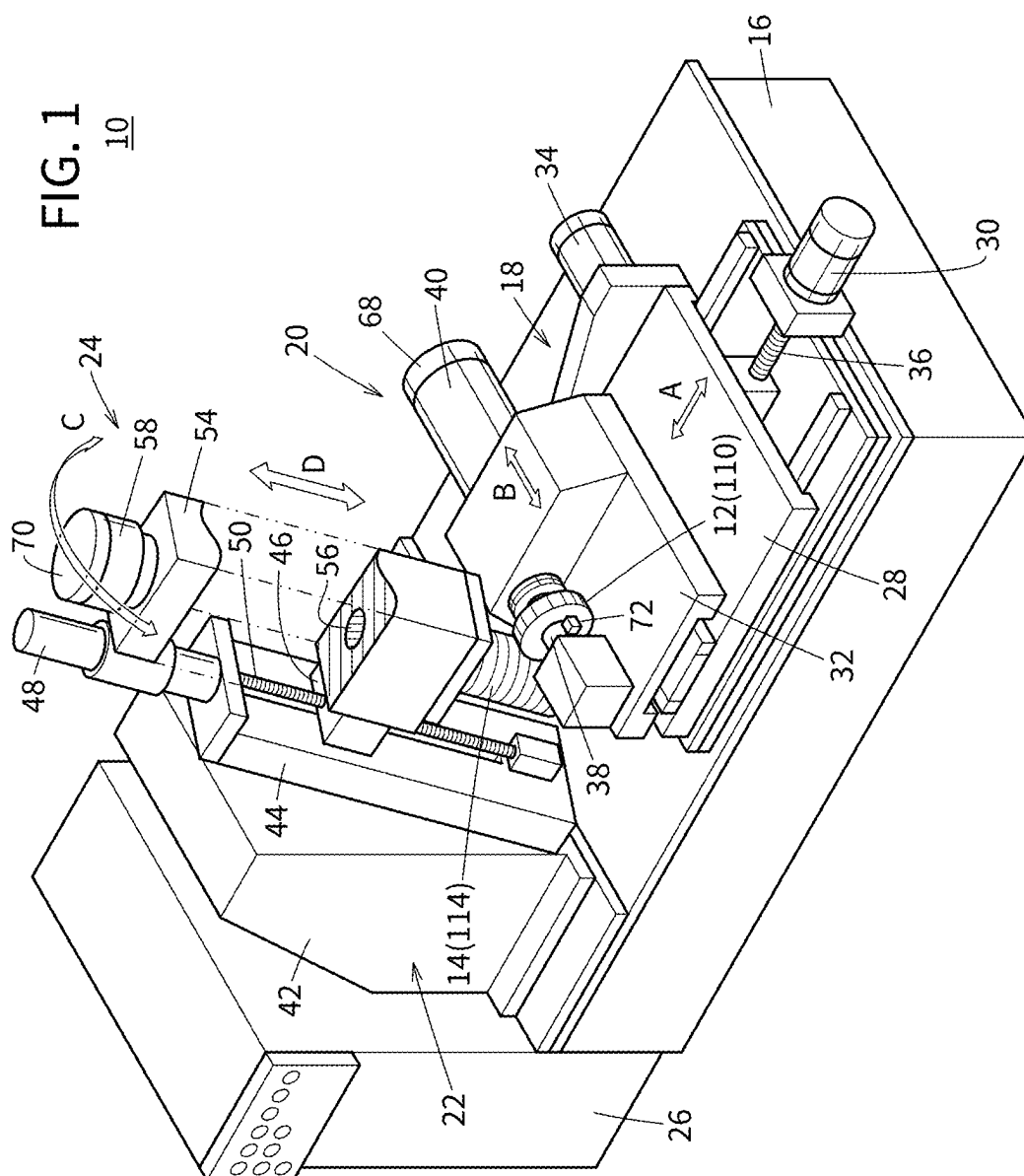


FIG. 1



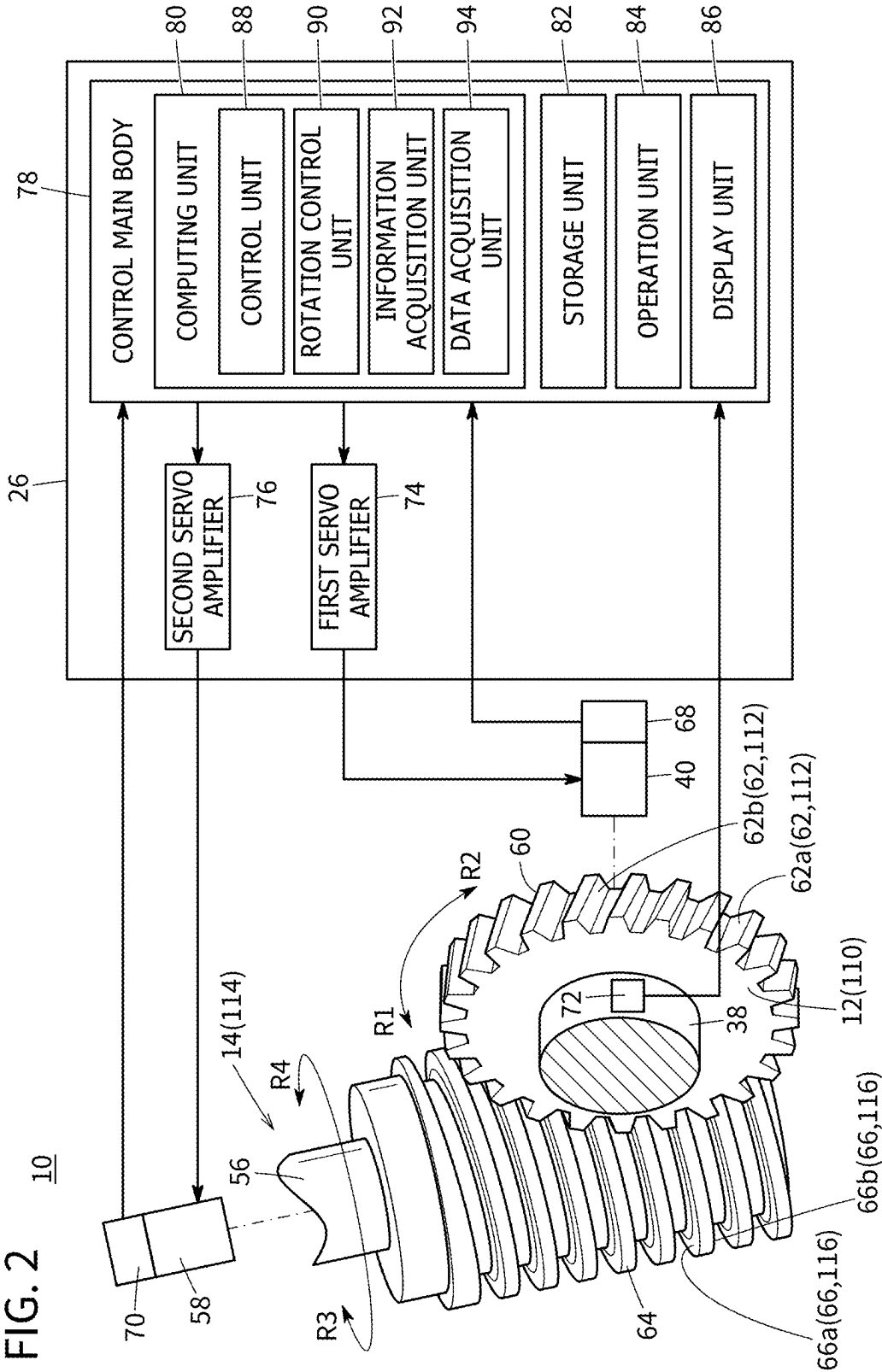


FIG. 3

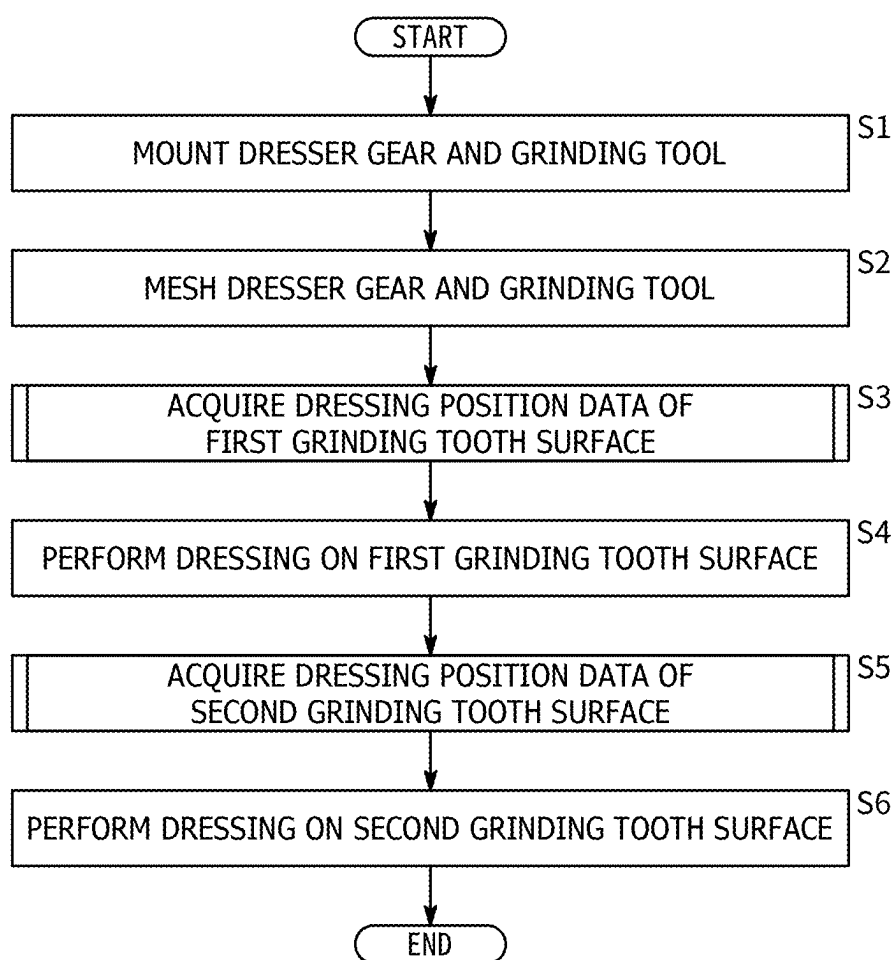


FIG. 4

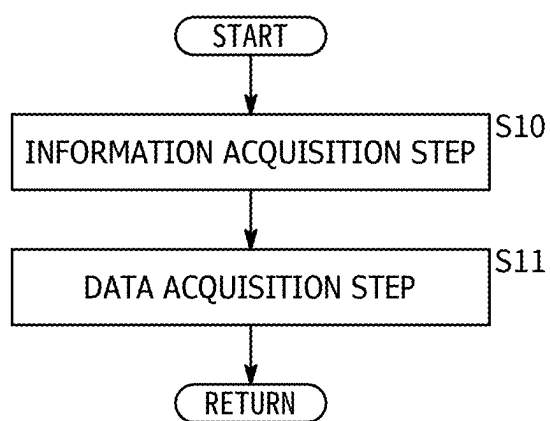


FIG. 5

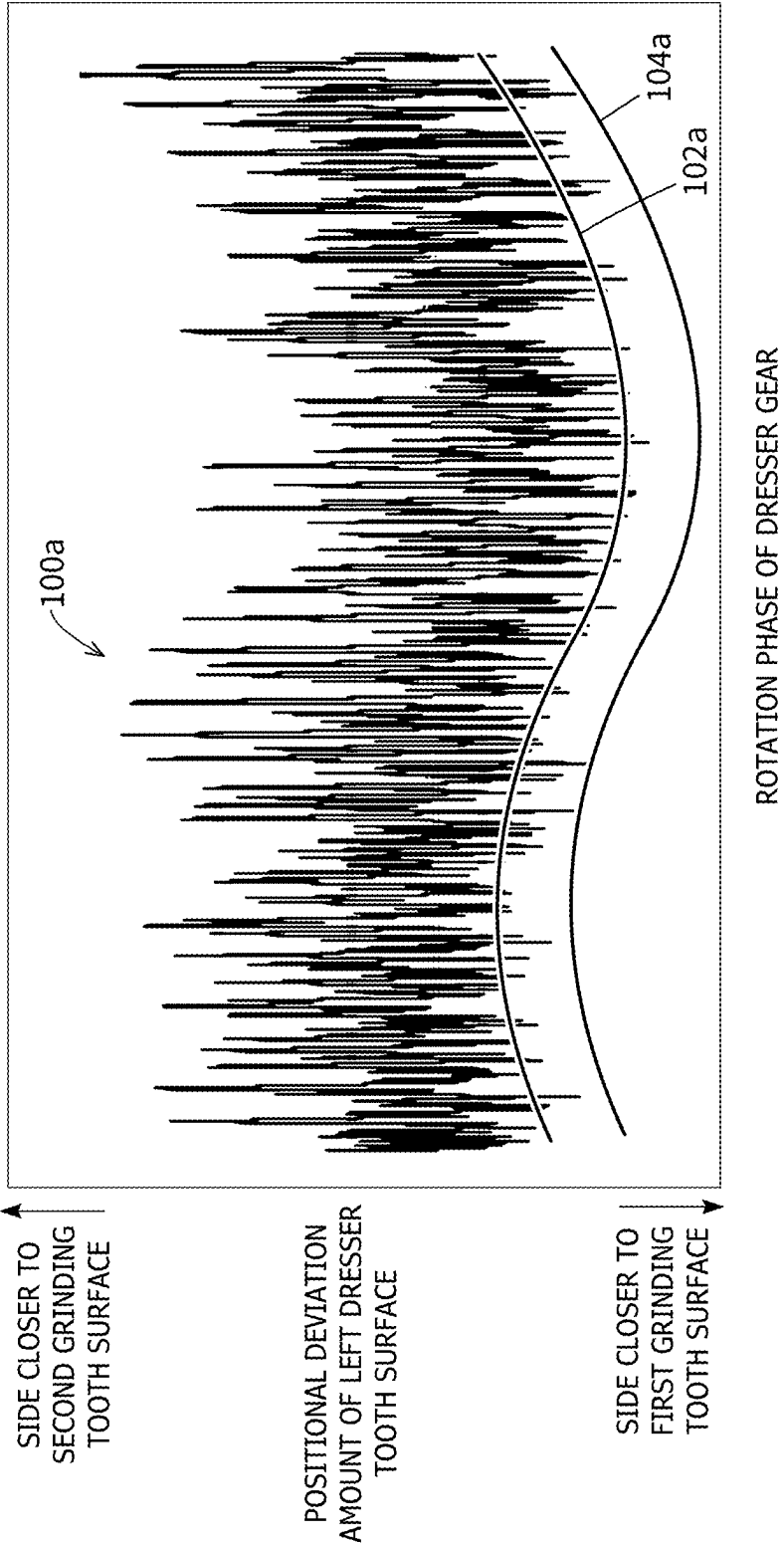
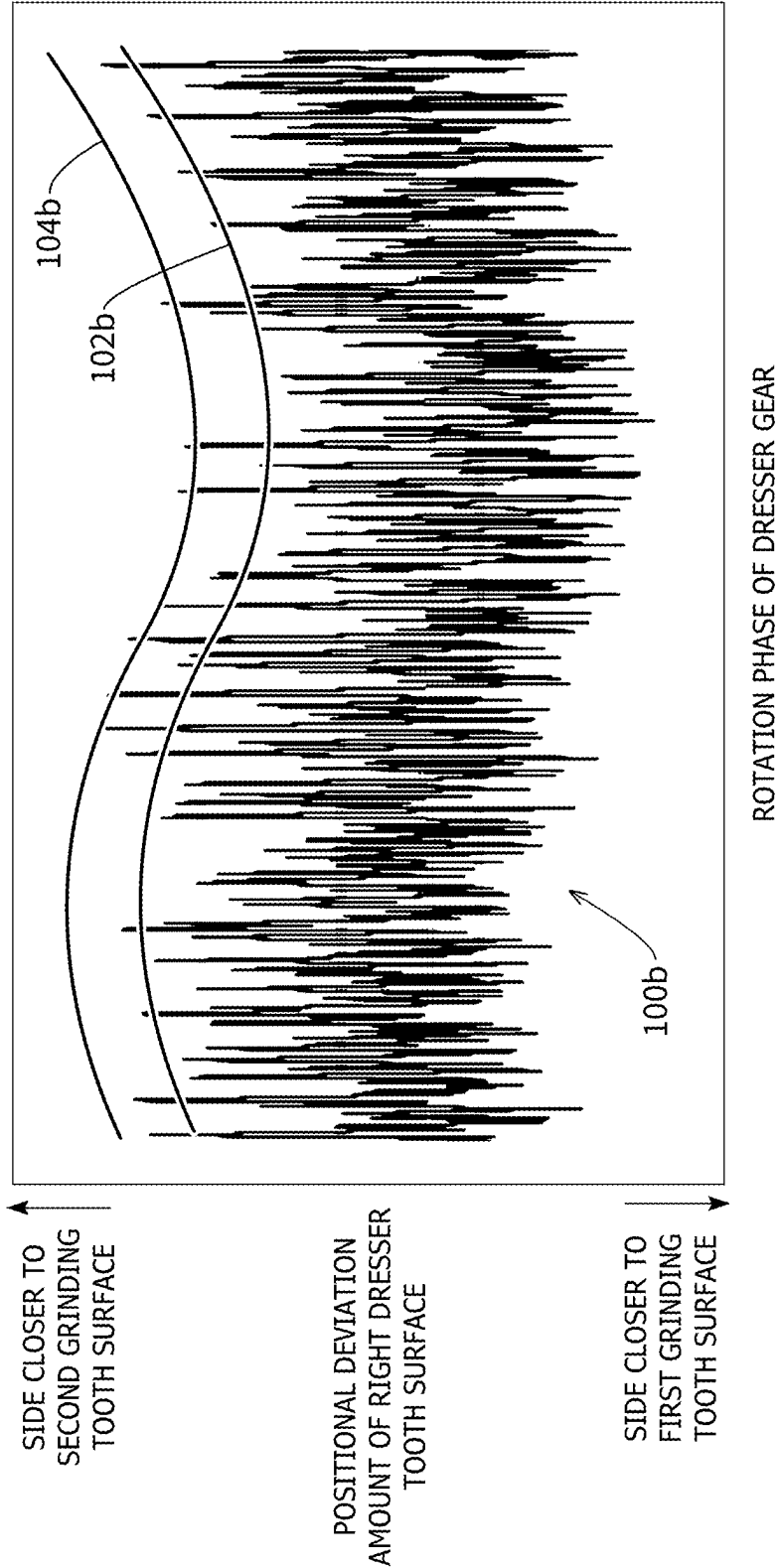


FIG. 6



## GRINDING TOOL DRESSING METHOD AND GRINDING DEVICE

### CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-037060 filed on Mar. 11, 2024, the contents of which are incorporated herein by reference.

### BACKGROUND OF THE INVENTION

#### Field of the Invention

[0002] The present disclosure relates to a grinding tool dressing method and a grinding device.

#### Description of the Related Art

[0003] JP 5367085 B2 discloses a method for dressing a grinding tool in which a grinding tool and a dresser gear are meshed and rotated, whereby a helical grinding tooth surface of the grinding tool is dressed by a dresser tooth surface of the dresser gear.

### SUMMARY OF THE INVENTION

[0004] Better grinding tool dressing methods and grinding devices are long-awaited.

[0005] The present disclosure aims to solve the aforementioned problems.

[0006] A first aspect of the present disclosure is a grinding tool dressing method for dressing a helical grinding tooth surface of a grinding tool with a dresser tooth surface of a dresser gear, wherein one of the grinding tool or the dresser gear is a first rotating body including a first tooth surface that is one of the grinding tooth surface or the dresser tooth surface, the other of the grinding tool or the dresser gear is a second rotating body including a second tooth surface that is the other of the grinding tooth surface or the dresser tooth surface, the grinding tool dressing method comprising: acquiring information on a deflection waveform indicating a positional deviation of the first tooth surface for each rotation phase of the first rotating body while a rotation speed of the first rotating body is changed in a manner so that the first tooth surface and the second tooth surface are in contact with each other in a state where the first rotating body and the second rotating body are meshed with each other and rotated, acquiring, as contact positional data indicating a position where the first tooth surface and the second tooth surface are in contact with each other, an envelope of the acquired deflection waveform, the envelope passing through a plurality of crests of the deflection waveform located on a side closer to the second tooth surface, and dressing the grinding tooth surface with the dresser tooth surface by changing a rotation speed of the first rotating body based on a predetermined forming quantity and the contact positional data in a state where the first rotating body and the second rotating body are meshed and rotated.

[0007] A second aspect of the present disclosure is a grinding device configured to dress a helical grinding tooth surface of a grinding tool with the dresser tooth surface of a dresser gear, wherein one of the grinding tool or the dresser gear is a first rotating body including a first tooth surface that is one of the grinding tooth surface and the dresser tooth surface, the other of the grinding tool or the dresser gear is

a second rotating body having a second tooth surface that is the other of the grinding tooth surface or the dresser tooth surface, and the grinding device comprising a rotation control unit configured to change a rotation speed of the first rotating body in a manner so that the first tooth surface and the second tooth surface are in contact with each other in a state where the first rotating body and the second rotating body are meshed and rotated, an information acquisition unit configured to acquire information on a deflection waveform indicating a positional deviation of the first tooth surface for each rotation phase of the first rotating body, and a data acquisition unit configured to acquire, as contact positional data indicating a position where the first tooth surface and the second tooth surface are contact with each other, an envelope of the deflection waveform acquired by the information acquisition unit, the envelope passing through a plurality of crests of the deflection waveform on a side closer to the second tooth surface, wherein the rotation control unit dresses the grinding tooth surface with the dresser tooth surface by changing a rotation speed of the first rotating body based on a predetermined forming quantity and the contact positional data in a state where the first rotating body and the second rotating body are meshed and rotated.

[0008] According to the present disclosure, a better grinding tool dressing method and grinding device can be provided.

[0009] The above and other objects, features, and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings, in which a preferred embodiment of the present invention is shown by way of illustrative example.

### BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a perspective view of a grinding device according to an embodiment;

[0011] FIG. 2 is a control block diagram of the grinding device;

[0012] FIG. 3 is a flow chart illustrating one example of a grinding tool dressing method;

[0013] FIG. 4 is a flow chart showing an example of the grinding tool dressing method;

[0014] FIG. 5 is a graph showing deflection waveform of a left dresser tooth surface; and

[0015] FIG. 6 is a graph showing deflection waveform of a right dresser tooth surface.

### DETAILED DESCRIPTION OF THE INVENTION

[0016] In a grinding tool dressing method, if an axial misalignment occurs when a dresser gear is mounted on a gear mounting shaft of a grinding device, the dresser gear vibrates during rotation. In this case, there is a case that a grinding tooth surface cannot be dressed by a dresser tooth surface.

[0017] As a method for solving such a problem, for example, the following grinding tool dressing method is considered. The grinding tool dressing method includes, for example, a first step, a second step, and a dressing step. In the first step, with the grinding tool and the dresser gear meshed and rotated, a deflection waveform that indicates a positional deviation of the dresser tooth surface for each rotation phase of the dresser gear is acquired while the



rotation speed of the dresser gear is changed in a manner so that the grinding tooth surface and the dresser tooth surface come into contact with each other.

**[0018]** In the second step, a line passing through an average value of the positional deviations against the rotation phase of the dresser gear in the deflection waveform is acquired as reference positional data. In the second step, dressing positional data is acquired by adding a forming quantity (infeed data) to the reference positional data. In the dressing step, while the grinding tool and the dresser gear are meshed and rotated, the rotation speed of the dresser gear is changed based on the dressing positional data, whereby the grinding tooth surface is dressed by the dresser tooth surface. According to such a method, even when an axial misalignment occurs between, for example, the gear mounting shaft and the dresser gear, the grinding tooth surface can be dressed.

**[0019]** The dresser tooth surface has abrasive grains for dressing the grinding tooth surface. The abrasive grains are distributed in a nonuniform manner on the dresser tooth surface. The non-uniformity in the distribution of the abrasive grains on the dresser tooth surface appears as the inconsistencies in an amplitude of the deflection waveform. A circumferential speed of the dresser gear when the rotation speed of the dresser gear is changed increases as the outer diameter of the dresser gear increases. Therefore, the amplitude of the deflection waveform varies with the size and the shape of the dresser gear.

**[0020]** A method of acquiring the dressing positional data by adding the forming quantity to the reference positional data passing through the average value for each rotation phase of the dresser gear in the deflection waveform may fail to accurately dress the grinding tooth surface due to the influence of the inconsistencies in the amplitude of the deflection waveform.

**[0021]** The present disclosure can provide a grinding tool dressing method and a grinding device capable of accurately dressing a grinding tooth surface without being affected by the inconsistencies in the amplitude of the deflection waveform.

**[0022]** FIG. 1 is a perspective view of a grinding device 10 according to an embodiment. As shown in FIG. 1, the grinding device 10 is a device for dressing a grinding tool 14 with a dresser gear 12. The grinding tool 14 can grind a workpiece gear (not shown) using the dressed grinding tool 14.

**[0023]** As shown in FIG. 1, the grinding device 10 includes a bed 16, a gear support mechanism 18, a gear rotation mechanism 20, a tool support mechanism 22, a tool rotation mechanism 24, and a controller 26.

**[0024]** The bed 16 is placed, for example, on a horizontal surface of a factory or the like. The gear support mechanism 18 is disposed on a flat upper surface of the bed 16. The gear support mechanism 18 includes a cutting table 28, a cutting motor 30, a traverse table 32, and a traverse motor 34.

**[0025]** The cutting table 28 moves in the A direction with respect to the bed 16. The A direction is a horizontal direction perpendicular to the height direction of the bed 16. The cutting table 28 is connected to the cutting motor 30 via a ball screw shaft 36. The cutting motor 30 moves the cutting table 28 in the A direction by rotating the ball screw shaft 36.

**[0026]** The traverse table 32 is disposed on the upper surface of the cutting table 28. The traverse table 32 moves in the B direction with respect to the cutting table 28. The B

direction is a direction perpendicular to the height direction of the bed 16 and the A direction. The traverse table 32 is coupled to the traverse motor 34 via a ball screw shaft (not shown). The traverse motor 34 moves the traverse table 32 in the B direction by rotating the ball screw shaft.

**[0027]** The gear rotation mechanism 20 is arranged on an upper surface of the traverse table 32. The gear rotation mechanism 20 has a gear mounting shaft 38 and a first motor 40. The gear mounting shaft 38 extends in the B direction. The dresser gear 12 is attachable to and detachable from the gear mounting shaft 38. A work gear (not shown) can be mounted on the gear mounting shaft 38 in place of the dresser gear 12. The first motor 40 rotates the gear mounting shaft 38.

**[0028]** The tool support mechanism 22 includes a column 42, a pivot table 44, a shift table 46, and a shift motor 48. The column 42 is positioned on the upper surface of the bed 16 so as to face the gear support mechanism 18. The column 42 extends upward from the bed 16. The pivot table 44 is attached to a surface of the column 42 facing the gear support mechanism 18.

**[0029]** The pivot table 44 extends in one direction. A turning motor (not shown) turns the pivot table 44 in the C direction with respect to the column 42. The shift table 46 is provided on a surface of the pivot table 44 facing the gear support mechanism 18. The shift table 46 is coupled to the shift motor 48 via a ball screw shaft 50. The shift motor 48 is attached to the pivot table 44. The shift motor 48 moves the shift table 46 in the D direction with respect to the pivot table 44.

**[0030]** The tool rotation mechanism 24 includes a base 54, a tool mounting shaft 56, and a second motor 58. The base 54 is attached to a surface of the shift table 46 facing the gear support mechanism 18. The base 54 extends along the direction in which the pivot table 44 extends. The tool mounting shaft 56 is inserted through the base 54 along the direction in which the base 54 extends. The grinding tool 14 is attachable to and detachable from the tool mounting shaft 56. The second motor 58 rotates the tool mounting shaft 56.

**[0031]** As shown in FIG. 2, the dresser gear 12 is mounted on the gear mounting shaft 38. The dresser gear 12 can be rotated in the R1 direction and the R2 direction by the driving force of the first motor 40. The dresser gear 12 is a gear for dressing the grinding tool 14. The dresser gear 12 has a plurality of dresser teeth 60. Each of the plurality of dresser teeth 60 is formed with a dresser tooth surface 62. The dresser tooth surface 62 includes a left dresser tooth surface 62a and a right dresser tooth surface 62b. For example, diamond abrasive grains or the like are electrodeposited on the dresser tooth surface 62 via a nickel plating layer.

**[0032]** The grinding tool 14 is mounted on the tool mounting shaft 56. The grinding tool 14 can be rotated in the R3 and R4 directions by the driving force of the second motor 58. The grinding tool 14 is a tool for grinding a work gear (not shown). The grinding tool 14 has helical grinding teeth 64. Grinding tooth surfaces 66 are formed on the grinding teeth 64. The grinding tooth surface 66 includes a first grinding tooth surface 66a and a second grinding tooth surface 66b. For example, single-layer CBN (cubic boron nitride) abrasive grains or the like are electrodeposited on the grinding tooth surface 66 via a nickel plating layer.

**[0033]** When the grinding tool 14 is dressed by the dresser gear 12, the dresser gear 12 and the grinding tool 14 are

meshed with each other. With the dresser gear 12 and the grinding tool 14 meshed, the left dresser tooth surface 62a faces the first grinding tooth surface 66a, and the right dresser tooth surface 62b faces the second grinding tooth surface 66b. With the dresser gear 12 and the grinding tool 14 meshed with each other, the dresser gear 12 is rotated in the R1 direction and the grinding tool 14 is rotated in the R3 direction, whereby the first grinding tooth surface 66a can be dressed by the left dresser tooth surface 62a. With the dresser gear 12 and the grinding tool 14 meshed with each other, the dresser gear 12 is rotated in the R2 direction and the grinding tool 14 is rotated in the R4 direction, whereby the second grinding tooth surface 66b can be dressed by the right dresser tooth surface 62b.

[0034] The grinding device 10 further includes a first encoder 68, a second encoder 70, and a contact sensor 72. The first motor 40 is provided with the first encoder 68. The first encoder 68 outputs to the controller 26 information (e.g., pulse signals) on the rotation phase (rotation speed, rotation angle, rotation position, rotation amount) of the dresser gear 12.

[0035] The second motor 58 is provided with the second encoder 70. The second encoder 70 outputs to the controller 26 information (e.g., pulse signals) on the rotation phase (rotation speed, rotation angle, rotation position, and rotation amount) of the grinding tool 14.

[0036] The contact sensor 72 detects the contact between the dresser tooth surface 62 and the grinding tooth surface 66. The contact sensor 72 is mounted on a bearing (not shown) that supports the gear mounting shaft 38 in a rotatable manner. The contact sensor 72 is, for example, an AE (Acoustic Emission) sensor. The AE sensor detects an elastic wave (contact sound) generated when the dresser tooth surface 62 and the grinding tooth surface 66 come into contact with each other. The contact sensor 72 is not limited to the AE sensor. The contact sensor 72 may be, for example, a vibration sensor, a torque sensor, etc. The contact between the grinding tool 14 and the dresser gear 12 may be detected based on the accumulated pulse described in JP 3910427 B2.

[0037] The controller 26 includes a first servo amplifier 74, a second servo amplifier 76, and a control main body 78. The first servo amplifier 74 controls the rotation of the first motor 40 based on signals output from the control main body 78. The second servo amplifier 76 controls the rotation of the second motor 58 based on signals output from the control main body 78.

[0038] The control main body 78 includes a computing unit 80, a storage unit 82, an operation unit 84, and a display unit 86. The computing unit 80 is composed of a processor such as a CPU (Central Processing Unit) or a GPU (Graphics Processing Unit). That is, the computing unit 80 is formed by processing circuitry.

[0039] The computing unit 80 includes a control unit 88, a rotation control unit 90, an information acquisition unit 92, and a data acquisition unit 94. The control unit 88 controls the cutting motor 30, the traverse motor 34, a turning motor (not shown), and the shift motor 48. The rotation control unit 90 controls the rotation of the dresser gear 12 via the first servo amplifier 74. The rotation control unit 90 controls the rotation of the grinding tool 14 via the second servo amplifier 76.

[0040] The control unit 88, the rotation control unit 90, the information acquisition unit 92, and the data acquisition unit 94 can be realized by the computing unit 80 executing

programs stored in the storage unit 82. At least part of the control unit 88, the rotation control unit 90, the information acquisition unit 92, and the data acquisition unit 94 may be realized by an integrated circuit such as an ASIC (Application Specific Integrated Circuit), an FPGA (Field-Programmable Gate Array), or the like. In addition, at least part of the control unit 88, the rotation control unit 90, the information acquisition unit 92, and the data acquisition unit 94 may be configured by an electronic circuit including discrete devices.

[0041] The storage unit 82 is composed of a volatile memory (not shown) and a nonvolatile memory (not shown). Examples of the volatile memory include, for example, a RAM (Random Access Memory) or the like. The volatile memory is used as working memory of a processor to temporarily store data or the like required for processing or computing operations. Examples of the nonvolatile memory include, for example, a ROM (Read Only Memory), a flash memory, or the like. The non-volatile memory is used as memory for storage, storing programs, tables, maps, etc. At least part of the storage unit 82 may be provided in the above-described processor, integrated circuit, etc.

[0042] The operation unit 84 is used when a user operates the controller 26. The operation unit 84 may include a keyboard, a mouse, and the like. The display unit 86 is provided with a display element (not shown). As the display element, for example, a liquid crystal display element, an organic electroluminescence display element, or the like is used. The operation unit 84 and the display unit 86 may be configured by a touch panel (not shown) provided with such a display element.

[0043] Next, an example of a method for dressing the grinding tool 14 will be described. FIGS. 3 and 4 are flow charts illustrating one example of a method for dressing the grinding tool 14.

[0044] In step S1, the dresser gear 12 is mounted on the gear mounting shaft 38 and the grinding tool 14 is mounted on the tool mounting shaft 56. Thereafter, the process transitions to step S2.

[0045] In step S2, the dresser gear 12 and the grinding tool 14 are meshed with each other. Specifically, the control unit 88 controls the cutting motor 30, the traverse motor 34, the turning motor (not shown), and the shift motor 48 to mesh the dresser gear 12 and the grinding tool 14 with each other. Thereafter, the process transitions to step S3.

[0046] In step S3, dressing positional data 104a (see FIG. 5) of the first grinding tooth surface 66a is acquired. That is, in step S10 of FIG. 4, an information acquisition step is performed. In the information acquisition step, in a state where the dresser gear 12 (first rotating body 110) and the grinding tool 14 (second rotating body 114) are meshed with each other and rotated, information on a deflection waveform 100a (see FIG. 5) indicating the positional deviation of the left dresser tooth surface 62a for each rotation phase of the dresser gear 12 is acquired while the rotation speed of the dresser gear 12 is changed in a manner so that the left dresser tooth surface 62a (first tooth surface 112) and the first grinding tooth surface 66a (second tooth surface 116) are in contact with each other.

[0047] Specifically, the rotation control unit 90 controls the first motor 40 via the first servo amplifier 74 to rotate the dresser gear 12 in the R1 direction. The rotation control unit 90 controls the second motor 58 via the second servo

amplifier 76 to rotate the grinding tool 14 in the R3 direction. The rotation control unit 90 controls the first servo amplifier 74 and the second servo amplifier 76 using feedback in a manner so that the dresser gear 12 and the grinding tool 14 rotate maintaining the engaged state, based on the information output from the first encoder 68 and the information output from the second encoder 70.

[0048] For example, when an axial misalignment occurs between the dresser gear 12 and the gear mounting shaft 38, the left dresser tooth surface 62a of the dresser gear 12 does not stably contact the first grinding tooth surface 66a of the grinding tool 14. In other words, when the dresser gear 12 and the grinding tool 14 are meshed and rotated, the left dresser tooth surface 62a is in contact with and separated from the first grinding tooth surface 66a. Therefore, in this embodiment, the rotation control unit 90 changes the rotation speed of the dresser gear 12 in a manner so that the left dresser tooth surface 62a and the first grinding tooth surface 66a come into contact with each other. The rotation control unit 90 rotates the grinding tool 14 at a predetermined constant rotation speed.

[0049] The contact sensor 72 detects elastic waves generated by the left dresser tooth surface 62a and the first grinding tooth surface 66a coming into contact with each other and outputs the detected elastic waves to the control main body 78. The control unit 88 determines whether the magnitude of the elastic waves is within a predetermined contact range. For example, the lower limit of the contact range can be set to a value slightly larger than elastic waves in a state where the left dresser tooth surface 62a and the first grinding tooth surface 66a are not in contact with each other. For example, the upper limit of the contact range can be set to a value slightly smaller than the elastic waves in a state where the left dresser tooth surface 62a is excessively in contact with the first grinding tooth surface 66a. That is, if the magnitude of the elastic waves output from the contact sensor 72 is within the contact range, the left dresser tooth surface 62a and the first grinding tooth surface 66a are in contact with each other in an appropriate manner. The contact range can be set as appropriate.

[0050] When the control unit 88 determines that the magnitude of the elastic waves output from the contact sensor 72 is smaller than the lower limit of the contact range, the rotation control unit 90 increases the rotation speed of the dresser gear 12. Thus, the left dresser tooth surface 62a moves in the direction towards the first grinding tooth surface 66a, so that the left dresser tooth surface 62a and the first grinding tooth surface 66a can be brought into contact with each other in an appropriate manner.

[0051] When the control unit 88 determines that the magnitude of the elastic waves output from the contact sensor 72 is larger than the upper limit value of the contact range, the rotation control unit 90 reduces the rotation speed of the dresser gear 12. Thus, the left dresser tooth surface 62a moves in the direction away from the first grinding tooth surface 66a, so that the left dresser tooth surface 62a and the first grinding tooth surface 66a can be brought into contact with each other in an appropriate manner.

[0052] Thus, when the rotational speed of the dresser gear 12 is changed in a manner so that the left dresser tooth surface 62a and the first grinding tooth surface 66a come into contact with each other, the deflection of the left dresser tooth surface 62a is reflected in the information output from the first encoder 68. Therefore, the information acquisition

unit 92 can acquire information on the deflection waveform 100a indicating the positional deviation of the dresser tooth surface 62 for each rotation phase of the dresser gear 12, based on the information output from the first encoder 68.

[0053] FIG. 5 is a graph showing the deflection waveform 100a of the left dresser tooth surface 62a. In the graph of FIG. 5, the horizontal axis indicates the rotation phase of the dresser gear 12, and the vertical axis indicates a positional deviation amount of the left dresser tooth surface 62a. In this embodiment, the information acquisition unit 92 acquires information on the deflection waveform 100a over one rotation of the dresser gear 12. After this, the rotation of the dresser gear 12 and the grinding tool 14 is temporarily stopped, and the process shifts to step S11.

[0054] In step S11, a data acquisition step is performed. In the data acquisition step, as shown in FIG. 5, an envelope passing through multiple crests on a side of the deflection waveform 100a acquired in the information acquisition step, the side being closer to the first grinding tooth surface 66a, are acquired as contact positional data 102a indicating positions at which the left dresser tooth surface 62a and the first grinding tooth surface 66a are in contact with each other.

[0055] In the data acquisition step, the rotational phase of the dresser gear 12 may be divided into a plurality of sections (e.g., 128 sections) in the deflection waveform 100a; the positional deviation amount at the position where the phase is the smallest may be extracted as a representative point from each section; and the contact positional data 102a may be acquired based on these representative points. In this case, since the load of data processing is reduced, the contact positional data 102a can be acquired in a relatively short time.

[0056] In the data acquisition step, dressing positional data 104a is acquired in which a predetermined forming quantity is added to the contact positional data 102a. Specifically, the contact positional data 102a shown in FIG. 5 is slid toward a side closer to the first grinding tooth surface 66a by the forming quantity, whereby the dressing positional data 104a of the first grinding tooth surface 66a is acquired. The forming quantity is machining allowance for the abrasive grains of the first grinding tooth surface 66a and is set as appropriate. The forming quantity is stored in advance in the storage unit 82. After this, the process transitions to step S4 (dressing step) in FIG. 3.

[0057] In step S4, the first grinding tooth surface 66a is dressed. Specifically, in a state where the dresser gear 12 and the grinding tool 14 are meshed and rotated, the rotation speed of the dresser gear 12 is changed based on the dressing positional data 104a of the first grinding tooth surface 66a, whereby the first grinding tooth surface 66a is dressed by the left dresser tooth surface 62a. The dressing of the first grinding tooth surface 66a is performed over the circumference of the dresser gear 12. Thereafter, the process transitions to step S5.

[0058] In step S5, dressing positional data 104b (see FIG. 6) of the second grinding tooth surface 66b is acquired. The dressing positional data 104b of the second grinding tooth surface 66b may be acquired by performing the processing of steps S10 and S11 described above between the right dresser tooth surface 62b and the second grinding tooth surface 66b. The description of the portion overlapping with the description of the acquisition of the dressing positional data 104a of the first grinding tooth surface 66a is omitted.

[0059] That is, in this case, in step S10, in the information acquisition step, in a state where the dresser gear 12 and the grinding tool 14 are meshed with each other and rotated, the information on the deflection waveform 100b (see FIG. 6) indicating a positional deviation of the right dresser tooth surface 62b for each rotation phase of the dresser gear 12 is acquired while the rotation speed of the dresser gear 12 is changed in a manner so that the right dresser tooth surface 62b (first tooth surface 112) and the second grinding tooth surface 66b (second tooth surface 116) are in contact with each other. FIG. 6 is a graph showing the deflection waveform of the right dresser tooth surface 62b.

[0060] In step S11, as shown in FIG. 6, an envelope passing through multiple crests on a side of the deflection waveform 100b of the right dresser tooth surface 62b, the side closer to the second grinding tooth surface 66b, are acquired as the contact positional data 102b indicating positions at which the right dresser tooth surface 62b and the second grinding tooth surface 66b are in contact with each other. Further, the contact positional data 102b shown in FIG. 6 is slid toward a side closer to the second grinding tooth surface 66b by the forming quantity, whereby the dressing positional data 104b of the second grinding tooth surface 66b is obtained. The forming quantity in this case is machining allowance for the abrasive grains of the second grinding tooth surface 66b and is set as appropriate. Then, the process transitions to step S6 (dressing step).

[0061] In step S6, the second grinding tooth surface 66b is dressed. Specifically, with the dresser gear 12 and the grinding tool 14 meshed and rotated, the rotation speed of the dresser gear 12 is changed based on the dressing positional data 104b of the second grinding tooth surface 66b, whereby the second grinding tooth surface 66b is dressed by the right dresser tooth surface 62b. The dressing of the second grinding tooth surface 66b is performed over the circumference of the dresser gear 12. After this, the process of FIG. 3 is completed.

[0062] In this embodiment, the dressing positional data 104a, 104b need not be acquired after the contact positional data 102a, 102b are acquired in the data acquisition step. In this case, in the dressing step, for example, the rotation control unit 90 may change the rotation speed of the dresser gear 12 by separately referring to the contact positional data 102a, 102b and the forming quantity.

[0063] According to the present embodiment, the envelopes passing through the multiple crests on sides of the deflection waveforms 100a, 100b of the dresser tooth surface 62, the sides closer to the grinding tooth surface 66, are acquired as the contact positional data 102a, 102b indicating positions where the dresser tooth surface 62 and the grinding tooth surface 66 are in contact with each other. Thus, the contact positional data 102a, 102b (reference position) between the dresser tooth surface 62 and the grinding tooth surface 66 can be grasped with high accuracy. That is, the contact positional data 102a, 102b are not affected by the inconsistencies in the amplitude of the deflection waveforms 100a, 100b. The dresser gear 12 is changed based on the predetermined forming quantity and the contact positional data 102a, 102b, whereby the grinding tooth surface 66 is dressed by the dresser tooth surface 62. Thus, the grinding tooth surface 66 can be dressed with high accuracy without being affected by the inconsistencies in the amplitude of the

deflection waveforms 100a, 100b. Thus, a better method for dressing the grinding tool 14 and grinding device 10 can be provided.

[0064] In the above-described embodiment, the dresser gear 12 is a first rotating body 110 having a first tooth surface 112, which is the dresser tooth surface 62. The grinding tool 14 is a second rotating body 114 having a second tooth surface 116, which is the grinding tooth surface 66. The present disclosure is not limited to such a configuration, and for example, the grinding tool 14 may be the first rotating body 110 and the dresser gear 12 may be the second rotating body 114. In this case, in the information acquisition step, in a state where the grinding tool 14 (first rotating body 110) and the dresser gear 12 (second rotating body 114) are meshed with each other and rotated, information on the deflection waveforms 100a, 100b indicating the positional deviation of the grinding tooth surface 66 for each rotation phase of the grinding tool 14 is acquired while the rotation speed of the grinding tool 14 is changed in a manner so that the grinding tooth surface 66 (first tooth surface 112) and the dresser tooth surface 62 (second tooth surface 116) come into contact with each other. In the data acquisition step, the envelopes passing through the multiple crests on the sides of the deflection waveforms 100a, 100b acquired in the information acquisition step, the sides closer to the grinding tooth surface 66, are acquired as the contact positional data 102a, 102b indicating positions the dresser tooth surface 62 and the grinding tooth surface 66 are in contact with each other. Furthermore, in the dressing step, while the dresser gear 12 and the grinding tool 14 are meshed and rotated, the rotation speed of the grinding tool 14 is changed based on the predetermined forming quantity and the contact positional data 102a, 102b, whereby the grinding tooth surface 66 is dressed by the dresser tooth surface 62.

[0065] With respect to the above embodiments, the following supplementary notes are further disclosed.

(Supplementary Note 1)

[0066] A method of the present disclosure for dressing a grinding tool (14) is a grinding tool dressing method for dressing a helical grinding tooth surface (66) of the grinding tool with a dresser tooth surface (62) of a dresser gear (12), wherein one of the grinding tool or the dresser gear is a first rotating body (110) including a first tooth surface (112) that is one of the grinding tooth surface or the dresser tooth surface, the other of the grinding tool or the dresser gear is a second rotating body (114) including a second tooth surface (116) that is the other of the grinding tooth surface or the dresser tooth surface, the grinding tool dressing method comprising: acquiring information on a deflection waveform (100a, 100b) indicating a positional deviation of the first tooth surface for each rotation phase of the first rotating body while a rotation speed of the first rotating body is changed in a manner so that the first tooth surface and the second tooth surface are in contact with each other in a state where the first rotating body and the second rotating body are meshed with each other and rotated, acquiring, as contact positional data (102a, 102b) indicating a position where the first tooth surface and the second tooth surface are in contact with each other, an envelope of the acquired deflection waveform, the envelope passing through a plurality of crests of the deflection waveform located on a side closer to the second tooth surface, and dressing the grinding tooth surface with the dresser tooth surface by changing a rotation speed

of the first rotating body based on a predetermined forming quantity and the contact positional data in a state where the first rotating body and the second rotating body are meshed and rotated.

[0067] According to such a method, an envelope passing through multiple crests located on a side closer to the second tooth surface among the deflection waveform of the first tooth surface are acquired as the contact positional data indicating a position where the first tooth surface and the second tooth surface are in contact with each other. Thus, the contact positional data (reference position) between the first and second tooth surfaces can be grasped with high accuracy. That is, the contact positional data are not affected by the inconsistencies in the amplitude of the deflection waveforms. The first gear is changed based on the predetermined forming quantity and the contact positional data, whereby the grinding tooth surface is dressed by the dresser tooth surface. Thus, the grinding tooth surface can be dressed with high accuracy without being affected by the inconsistencies in the amplitude of the deflection waveforms. Thus, a better method for dressing the grinding tool can be provided.

(Supplementary Note 2)

[0068] Regarding the grinding tool dressing method according to Supplementary note 1, the first tooth surface of the first rotating body may be the dresser tooth surface of the dresser gear, and the second tooth surface of the second rotating body may be the grinding tooth surface of the grinding tool.

(Supplementary Note 3)

[0069] Regarding the grinding tool dressing method according to Supplementary note 1 or 2, in the acquiring of the envelop, dressing positional data (104a, 104b) is acquired in which the forming quantity is added to the contact positional data; and in the dressing, the grinding tooth surface is dressed with the dresser tooth surface by changing a rotation speed of the first rotating body based on the dressing positional data in a state where the first rotating body and the second rotating body are meshed and rotated.

[0070] According to such a method, it is possible to dress the grinding tooth surface with the dresser tooth surface by simple control by acquiring the dressing positional data.

(Supplementary Note 4)

[0071] A grinding device of the present disclosure is a grinding device (10) configured to dress a helical grinding tooth surface of a grinding tool with the dresser tooth surface of a dresser gear, wherein one of the grinding tool or the dresser gear is a first rotating body including a first tooth surface that is one of the grinding tooth surface and the dresser tooth surface, the other of the grinding tool or the dresser gear is a second rotating body having a second tooth surface that is the other of the grinding tooth surface or the dresser tooth surface, and the grinding device comprising a rotation control unit (90) configured to change a rotation speed of the first rotating body in a manner so that the first tooth surface and the second tooth surface are in contact with each other in a state where the first rotating body and the second rotating body are meshed and rotated, an information acquisition unit (92) configured to acquire information on a deflection waveform indicating a positional deviation of the first tooth surface for each rotation phase of the first rotating

body, and a data acquisition unit (94) configured to acquire, as contact positional data indicating a position where the first tooth surface and the second tooth surface are contact with each other, an envelope of the deflection waveform acquired by the information acquisition unit, the envelope passing through a plurality of crests of the deflection waveform on a side closer to the second tooth surface, wherein the rotation control unit dresses the grinding tooth surface with the dresser tooth surface by changing a rotation speed of the first rotating body based on a predetermined forming quantity and the contact positional data in a state where the first rotating body and the second rotating body are meshed and rotated.

[0072] According to such a configuration, a grinding device that achieves the same effect as that of Supplementary note 1 can be obtained. Thus, a better grinding device can be provided.

(Supplementary Note 5)

[0073] Regarding the grinding device according to Supplementary note 4, the first tooth surface of the first rotating body may be the dresser tooth surface of the dresser gear, and the second tooth surface of the second rotating body may be the grinding tooth surface of the grinding tool.

(Supplementary Note 6)

[0074] Regarding the grinding device according to supplementary note 4 or 5, the data acquisition unit may acquire dressing positional data in which the forming quantity is added to the contact positional data, and the rotation control unit may dress the grinding tooth surface by the dresser tooth surface by changing the rotation speed of the first rotating body based on the dressing positional data in a state where the first rotating body and the second rotating body are meshed and rotated.

[0075] Although the present disclosure has been detailed, the present disclosure is not limited to the individual embodiments described above. These embodiments may be variously added, replaced, altered, partially deleted, etc., without departing from the scope of the present disclosure or the intent of the present disclosure as derived from the claims and their equivalents. These embodiments can also be implemented in combination. For example, in the above-described embodiment, the order of the operations and the order of the processes are shown as an example, and are not limited to these. The same applies to the case where numerical values or mathematical expressions are used in the description of the above-described embodiment.

1. A grinding tool dressing method for dressing a helical grinding tooth surface of a grinding tool with a dresser tooth surface of a dresser gear, wherein

one of the grinding tool or the dresser gear is a first rotating body including a first tooth surface that is one of the grinding tooth surface or the dresser tooth surface,

another of the grinding tool or the dresser gear is a second rotating body including a second tooth surface that is the other of the grinding tooth surface or the dresser tooth surface,

the grinding tool dressing method comprising:

acquiring information on a deflection waveform indicating a positional deviation of the first tooth surface for each rotation phase of the first rotating body while a

- rotation speed of the first rotating body is changed in a manner so that the first tooth surface and the second tooth surface are in contact with each other in a state where the first rotating body and the second rotating body are meshed and rotated;
- acquiring, as contact positional data indicating a position where the first tooth surface and the second tooth surface are in contact with each other, an envelope of the acquired deflection waveform, the envelop passing through a plurality of crests of the deflection waveform located on a side closer to the second tooth surface; and
- dressing the grinding tooth surface with the dresser tooth surface by changing the rotation speed of the first rotating body based on a predetermined forming quantity and the contact positional data in the state where the first rotating body and the second rotating body are meshed and rotated.
2. The grinding tool dressing method according to claim 1, wherein
- the first tooth surface of the first rotating body is the dresser tooth surface of the dresser gear, and
- the second tooth surface of the second rotating body is the grinding tooth surface of the grinding tool.
3. The grinding tool dressing method according to claim 1, further comprising:
- acquiring dressing positional data in which the forming quantity is added to the contact positional data; and
- dressing the grinding tooth surface with the dresser tooth surface by changing the rotation speed of the first rotating body based on the dressing positional data in the state where the first rotating body and the second rotating body are meshed and rotated.
4. A grinding device configured to dress a helical grinding tooth surface of a grinding tool with a dresser tooth surface of a dresser gear, wherein
- one of the grinding tool or the dresser gear is a first rotating body including a first tooth surface that is one of the grinding tooth surface or the dresser tooth surface,
- the other of the grinding tool or the dresser gear is a second rotating body including a second tooth surface that is the other of the grinding tooth surface or the dresser tooth surface,

- the grinding device comprising a controller including one or more processors executing computer-executable instructions stored in memory,
- wherein the one or more processors execute the computer-executable instructions to cause the controller to:
- change a rotation speed of the first rotating body in a manner so that the first tooth surface and the second tooth surface are in contact with each other in a state where the first rotating body and the second rotating body are meshed and rotated;
- acquire information on a deflection waveform indicating a positional deviation of the first tooth surface for each rotation phase of the first rotating body;
- acquire, as contact positional data indicating a position where the first tooth surface and the second tooth surface are in contact with each other, an envelope of the acquired deflection waveform, the envelop passing through a plurality of crests of the deflection waveform located on a side closer to the second tooth surface, and
- dress the grinding tooth surface with the dresser tooth surface by changing the rotation speed of the first rotating body based on a predetermined forming quantity and the contact positional data in the state where the first rotating body and the second rotating body are meshed and rotated.
5. The grinding device according to claim 4, wherein
- the first tooth surface of the first rotating body is the dresser tooth surface of the dresser gear, and
- the second tooth surface of the second rotating body is the grinding tooth surface of the grinding tool.
6. The grinding device according to claim 4, wherein
- the one or more processors execute the computer-executable instructions to cause the controller to:
- acquire dressing positional data in which the forming quantity is added to the contact positional data; and
- dress the grinding tooth surface with the dresser tooth surface by changing the rotation speed of the first rotating body based on the dressing positional data in the state where the first rotating body and the second rotating body are meshed and rotated.

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